

AWS CALL TRANSCRIPTION & SENTIMENT ANALYSIS

Project Report submitted in the partial fulfilment

Of

Bachelor Of Technology

In

Data Science

By

JEET SHOREY(J024)

Under the supervision of

Prof.Rashmi Patel

(Assistant Professor, Department in Data Science, MPSTME)

SVKM's NMIMS University

(Deemed-to-be University)



**MUKESH PATEL SCHOOL OF TECHNOLOGY
MANAGEMENT & ENGINEERING MPSTME**

Vile Parle (W),Mumbai-56

(2025-2026)

CERTIFICATE



This is to certify that the project entitled “AWS CALL TRANSCRIPTION & SENTIMENT ANALYSIS”, has been done by Mr.Jeet Shorey J024 under my guidance and supervision & has been submitted in partial fulfilment of the degree of Btech in Data Science of MPSTME, SVKM’s NMIMS (Deemed-to-be University), Mumbai, India.

Project mentor (Prof. Rashmi Patel)
(Internal Guide)

Examiner (Name and Signature)

Dr. Siba Panda
(HoD)

Date:

Place: Mumbai

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who contributed to the successful completion of this project.

First and foremost, I am deeply grateful to my Internal Faculty Mentor, Prof. Rashmi Patel, Department of Data Science, MPSTME, for her continuous guidance, invaluable feedback, and encouragement throughout the course of this project. Her deep academic insights into the latest paradigms of Artificial Intelligence encompassing Large Language Models (LLMs), Multimodal Deep Learning, Retrieval-Augmented Generation (RAG), and Agentic AI Workflows, alongside specialized areas like Automatic Speech Recognition (ASR), Natural Language Understanding (NLU), Explainable AI (XAI), and MLOps have been absolutely instrumental in providing the technical foundation and rigorous standards required for this work.

I would like to express my deepest gratitude and heartfelt appreciation to my Industry Mentor, Mr. Nilesh Pabuwal Sir, and the entire AI/ML team at Kotak Mahindra Prime Loans for providing me with this incredible opportunity to work on a real-world, production-grade AI system in a highly professional environment. Their constant guidance, encouragement, technical mentorship, and unwavering support throughout the internship played a significant role in enhancing both my technical expertise and professional confidence. The invaluable hands-on exposure to AWS cloud infrastructure, speech processing technologies, machine learning workflows, and financial compliance systems gave me insights far beyond classroom learning and helped me understand the practical implementation of AI/ML solutions in the banking industry. I am truly thankful for their trust, patience, and continuous motivation, which made this internship an exceptionally enriching, inspiring, and transformative learning experience for me.

I am grateful to the Head of Department and all the faculty members of the Department of Data Science at MPSTME for their academic support and for creating an environment that fosters innovation and applied learning.

I also wish to thank my classmates and fellow interns for the collaborative spirit and constructive discussions that enriched my understanding of the subject matter.

Finally, I am deeply indebted to my family for their unwavering support, patience, and motivation throughout this academic journey.

NAME	ROLL NO.	SAP ID
JEET SHOREY	J024	70092200086

ABSTRACT

The banking and financial services sector generates vast volumes of customer interaction data through call centres. Analysing these interactions manually is time consuming, error prone, and operationally expensive. This project presents the design, development, and deployment of an automated AI/ML pipeline for call transcription and sentiment analysis, built specifically for the operations at Kotak Mahindra Prime Loans.

The system is architected on Amazon Web Services (AWS) and is event driven: upon upload of an audio file (MP3, WAV, or similar format) to Amazon S3, an automated trigger invokes a Dockerized AWS Lambda function. This function leverages OpenAI's Whisper model for high-accuracy automatic speech recognition (ASR) and transcription, supporting both Hindi and English languages commonly used in Indian banking call centres. The transcribed text is then passed through a speaker diarization module that distinguishes between the Agent and the Customer, enabling role-specific analysis. Subsequently, VADER (Valence Aware Dictionary and sEntiment Reasoner) is applied to compute sentiment scores at both the segment and overall call level. A customer satisfaction score on a scale of 1-10 is derived from the sentiment metrics.

A critical compliance feature is the detection of flagged keywords pre-defined terms related to regulatory non-compliance, mis-selling, or abusive language that are automatically identified and logged. Results, including the record ID, overall sentiment classification, customer satisfaction score, diarized transcript, and compliance flags, are stored in Amazon DynamoDB for downstream reporting and analytics.

The system is containerized using Docker and deployed to AWS Lambda via Amazon ECR, ensuring portability and scalability. An automated deployment script (deploy.py) handles the complete provisioning of AWS resources including S3 buckets, DynamoDB tables, IAM roles, and ECR repositories.

This project also provides an overview of other AI/ML initiatives undertaken during the internship, including a Loan and Credit Card Approval Prediction system and an

AI-powered Chatbot with Salesforce integration and OTP verification capabilities. Evaluation results demonstrate that the pipeline achieves high transcription accuracy across code-switched Hindi-English conversations, reliable sentiment classification, and low-latency end-to-end processing, making it a robust solution for banking call centre analytics and regulatory compliance monitoring.

TABLE OF CONTENTS

Certificate	i
Acknowledgement.....	ii
Abstract.....	iii
Table of Contents.....	iv
List of Figures.....	v
List of Tables.....	vi
List of Abbreviations.....	vii
Chapter 1: Introduction.....	1
1.1 Background of the project topic.....	1
1.2 Motivation and scope of the report.....	1
1.3 Problem statement.....	2
1.4 Salient contributions.....	2
1.5 Organisation of the report.....	3
1.6 Summary of internship projects.....	3
Chapter 2: Literature Survey.....	4
2.1 Introduction to the overall domain.....	4
2.2 Exhaustive literature survey.....	4
2.2.1 Automatic speech recognition.....	4
2.2.2 Speaker diarization.....	5
2.2.3 Sentiment analysis and NLP.....	6
2.2.4 Cloud-native and serverless ML architectures.....	6
2.3 Research gap and problem identification.....	8
Chapter 3: Methodology and Implementation.....	10
3.1 System architecture.....	10
3.1.1 High-level architecture overview.....	10
3.1.2 Detailed AWS services mapping.....	10
3.2 Hardware and infrastructure description.....	11
3.3 Software description, flowcharts, and algorithms.....	12
3.3.1 lambda_function.py — core processing logic.....	12
3.3.2 Speaker diarization algorithm.....	14
3.3.3 Compliance and critical word detection.....	15
3.3.4 Dockerfile Lambda container architecture.....	15
3.3.5 deploy.py infrastructure automation.....	16
3.3.6 DynamoDB output schema.....	17
3.4 Additional internship projects.....	18

3.4.1 Loan and credit card approval prediction.....	18
3.4.2 AI chatbot with Salesforce integration and OTP verification.....	19
Chapter 4: Results and Analysis.....	21
4.1 Transcription performance results.....	22
4.2 Sentiment analysis accuracy.....	22
4.3 Compliance feature performance.....	22
4.4 System performance metrics.....	23
Chapter 5: Advantages, Limitations and Applications.....	25
5.1 Advantages.....	25
5.2 Limitations.....	25
5.3 Applications.....	26
Conclusion and Future Scope.....	28
Conclusion.....	28
Future scope.....	29
References.....	30
Appendix I: System flowcharts summary.....	31
A.1 Master pipeline flowchart.....	31
Appendix II: Soft code flowcharts.....	33
B.1 Master pipeline workflow.....	33
B.2 Sentiment analysis flowchart.....	33
B.3 Compliance monitoring flowchart.....	33
B.4 Deployment automation flowchart.....	34

LIST OF FIGURES

Fig No.	Figure Title	Page No.
Fig 3.1	High Level System Architecture S3 to Lambda to DynamoDB Pipeline	10
Fig 3.2	Detailed AWS Infrastructure and Data Flow Diagram	12
Fig 3.3	End-to-End Process Flowchart Audio Upload to Result Storage	13
Fig 3.4	Speaker Diarization Logic Agent vs. Customer Separation	14
Fig 3.5	Docker Container Architecture for AWS Lambda	16
Fig 3.6	Deployment Automation Flow using deploy.py	16
Fig 3.7	Salesforce Architechture	19
Fig 4.1	DynamoDB output	23
Fig A.1	Master Pipeline Flowchart	30

LIST OF TABLES

Table No.	Table Title	Page No.
Table 1.1	Summary of Internship Projects at Kotak Mahindra Prime Loans	3
Table 2.1	Comparative Literature Survey of Speech Recognition Systems	8
Table 3.1	AWS Services Used and Their Roles in the Pipeline	11
Table 3.2	Technical Stack Summary	12
Table 3.3	Critical / Compliance Keywords List (Sample)	15
Table 3.4	DynamoDB Output Schema	17
Table 4.1	Transcription Accuracy Results Language-wise Breakdown	20
Table 4.2	Sentiment Analysis Accuracy Metrics	21
Table 4.3	Lambda Summary	22

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
ASR	Automatic Speech Recognition
AWS	Amazon Web Services
S3	Simple Storage Service
ECR	Elastic Container Registry
IAM	Identity and Access Management
DynamoDB	Amazon DynamoDB (NoSQL Database Service)
NLP	Natural Language Processing
VADER	Valence Aware Dictionary and sEntiment Reasoner
API	Application Programming Interface
SDK	Software Development Kit
OTP	One-Time Password
CRM	Customer Relationship Management
FFmpeg	Fast Forward Moving Picture Experts Group (Audio/Video Framework)
JSON	JavaScript Object Notation
CPU	Central Processing Unit
GPU	Graphics Processing Unit
MPSTME	Mukesh Patel School of Technology Management & Engineering
NMIMS	Narsee Monjee Institute of Management Studies
RBI	Reserve Bank of India

Chapter 1: Introduction

1.1 Background of the Project Topic

The Indian banking sector is one of the most interaction-intensive industries, with millions of calls handled daily across loan servicing, customer support, and collections departments. Kotak Mahindra Prime Loans, a subsidiary of Kotak Mahindra Bank, manages a significant volume of calls related to personal loans, vehicle loans, and credit products. These calls carry immense informational value they reveal customer satisfaction, compliance adherence by agents, sales quality, and potential regulatory risks.

Traditionally, quality assurance of such calls involved manual sampling, where a small fraction of calls were reviewed by supervisors. This approach is neither scalable nor comprehensive. With advances in cloud computing, automatic speech recognition (ASR), and natural language processing (NLP), it has become feasible to build systems that automatically transcribe, analyse, and flag every call in near real-time.

The convergence of OpenAI Whisper (a state-of-the-art multilingual ASR model), VADER sentiment analysis, and serverless AWS architecture provides an ideal foundation for such a system. This project leverages all three to build a production ready pipeline suited to the linguistic and operational realities of an Indian banking call centre, where code switching between Hindi and English is the norm.

1.2 Motivation and Scope of the Report

The motivation for this project stems from a clear operational need identified at Kotak Mahindra Prime Loans during the internship. Key pain points observed were: inability to monitor 100% of customer calls for quality and compliance; no automated mechanism to detect policy violations or customer dissatisfaction in real time; manual transcription being slow and expensive and no structured analytics on agent-customer interaction data.

The scope of this report covers the complete design, development, and testing of an end-to-end automated pipeline. It includes the AWS cloud infrastructure, the AI models used, the containerization strategy, the compliance monitoring features, and a discussion of three other AI/ML projects undertaken during the internship: Loan & Credit Card Approval Prediction and AI Chatbot with Salesforce Integration.

1.3 Problem Statement

Kotak Mahindra Prime Loans requires an automated, scalable, and cost-effective system that can: (i) transcribe audio recordings of customer calls with high accuracy across Hindi and English; (ii) identify and separately analyse the speech of the agent versus the customer (speaker diarization); (iii) perform sentiment analysis and derive a customer satisfaction score; (iv) detect critical words indicative of compliance violations or negative customer experiences; and (v) store all structured outputs for downstream reporting and regulatory audit.

The challenge is compounded by the use of Indian accents, code-mixed language (Hinglish), noisy call centre audio, and the need for the system to be serverless and auto-scaling to handle variable daily call volumes.

1.4 Salient Contributions

1. Design and deployment of a fully serverless, event-driven AWS pipeline for call analytics.
2. Integration of OpenAI Whisper for multilingual (Hindi + English) transcription within a Docker Lambda environment.
3. Implementation of speaker diarization to separate Agent and Customer speech segments.
4. Real-time compliance monitoring via critical keyword detection flagging mechanism.
5. Automated multi-resource AWS provisioning via a Python deployment script (deploy.py).
6. Structured DynamoDB schema for audit-ready storage of call analytics results.

7. Additional contributions: Loan/Credit Card Approval ML model and Salesforce-integrated AI Chatbot with OTP verification.

1.5 Organisation of the Report

Chapter 1 provides an introduction, motivation, and problem statement. Chapter 2 surveys the relevant literature in speech recognition, sentiment analysis, and cloud based NLP pipelines. Chapter 3 presents the system architecture, technical implementation details, and all workflows. Chapter 4 presents results and analysis including transcription accuracy and sentiment metrics. Chapter 5 discusses advantages, limitations, and real-world applications. The report concludes with future scope and a complete reference list.

1.6 Summary of Internship Projects

During the internship at Kotak Mahindra Prime Loans, three distinct AI/ML projects were undertaken. Table 1.1 provides a summary.

Table 1.1: Summary of Internship Projects at Kotak Mahindra Prime Loans

Project	Technology Stack	Key Outcome
AWS Call Transcription & Sentiment Analysis (Primary)	AWS S3, Lambda, ECR, DynamoDB, Whisper AI, VADER, Docker, Python	Automated call analytics pipeline with compliance monitoring
Loan & Credit Card Approval Prediction	Python, Scikit-learn, Pandas, ML Classification Models	Predictive model for loan eligibility and credit card approval decisions
AI Chatbot with Salesforce Integration & OTP Verification	Python, Salesforce API, OTP framework, NLP	Intelligent chatbot for customer service with CRM integration and secure OTP based verification

Chapter 2: Literature Survey

2.1 Introduction to the Overall Domain

The intersection of Automatic Speech Recognition (ASR), Natural Language Processing (NLP), and cloud native serverless architectures represents one of the most rapidly evolving areas of applied AI. This chapter surveys the key academic and industrial works that form the theoretical and technical foundation of this project.

Automatic speech recognition has advanced significantly with the advent of deep learning. Traditional Hidden Markov Model (HMM) based systems [1] were largely supplanted by deep neural network approaches, and more recently by transformer based end-to-end models. Sentiment analysis, similarly, has evolved from lexicon based methods to deep learning classifiers, though lexicon approaches like VADER remain highly effective for short, conversational text [2].

In the banking and financial services domain, the adoption of conversational AI and speech analytics has grown substantially. Research by Karpagam et al. [3] demonstrated that automated call centre analytics could reduce quality assurance costs by over 60% while increasing coverage to 100% of calls. Studies in the Indian context specifically highlight the challenge of code-switched (Hindi-English) speech, which standard ASR systems handle poorly without multilingual training data [4].

2.2 Exhaustive Literature Survey

2.2.1 Automatic Speech Recognition

[1] Graves, A., Mohamed, A., & Hinton, G. (2013). 'Speech Recognition with Deep Recurrent Neural Networks.' ICASSP 2013. This seminal paper introduced the Connectionist Temporal Classification (CTC) loss for sequence-to-sequence learning in ASR, enabling end-to-end training without explicit alignment between input audio and output labels. The work achieved state-of-the-art results on TIMIT and laid the groundwork for modern ASR architectures.

[2] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). 'Robust Speech Recognition via Large-Scale Weak Supervision.' OpenAI Technical Report / ICML 2023. This paper introduces Whisper, the ASR model used in this project. Whisper is trained on 680,000 hours of multilingual and multitask supervised data from the web. The authors demonstrate that the scale of diverse training data leads to robustness across accents, background noise, and technical language. Crucially for this project, Whisper supports Hindi and English natively, making it ideal for Indian banking call centre audio.

[3] Chan, W., Jaitly, N., Le, Q. V., & Vinyals, O. (2016). 'Listen, Attend and Spell: A Neural Network for Large Vocabulary Conversational Speech Recognition.' ICASSP 2016. The Listen, Attend, and Spell (LAS) architecture introduced the attention mechanism to ASR. It directly learns to map audio sequences to character sequences, removing the need for pronunciation dictionaries. This work is foundational to understanding why modern ASR models like Whisper outperform traditional approaches.

2.2.2 Speaker Diarization

[4] Park, T. J., Kanda, N., Dimitriadis, D., Han, K. J., Watanabe, S., & Narayanan, S. (2022). 'A Review of Speaker Diarization: Recent Advances with Deep Learning.' Computer Speech & Language, Vol. 72. This comprehensive review covers the evolution of speaker diarization from Gaussian Mixture Model-based systems to modern deep learning approaches using x-vectors and clustering methods. The paper identifies open challenges in overlapping speech and domain mismatch, which are relevant to call centre audio.

[5] Bredin, H., & Laurent, A. (2021). 'End-to-End Speaker Segmentation for Overlap-Aware Resegmentation.' Interspeech 2021. This paper presents an end-to-end neural approach to speaker segmentation. It is particularly relevant to the diarization component of this project, which uses audio energy and silence detection heuristics combined with Whisper timestamps to approximate agent-customer turn separation.

2.2.3 Sentiment Analysis and NLP

[6] Hutto, C. J., & Gilbert, E. (2014). 'VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text.' ICWSM 2014. VADER is the primary sentiment analysis engine used in this project. The authors demonstrate that VADER outperforms individual human raters and is highly effective for short, conversational text exactly the type of utterances produced by speaker diarization. VADER's compound score, which ranges from -1 to +1, directly informs the customer satisfaction metric in this system.

[7] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). 'BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.' NAACL 2019. BERT represents the state of the art in contextual NLP. While this project uses VADER for efficiency and real-time processing within Lambda constraints, BERT based models are identified in the future scope as a superior alternative for nuanced sentiment classification in longer conversations.

2.2.4 Cloud-Native and Serverless ML Architectures

[8] Amazon Web Services. (2023). 'AWS Lambda: Serverless Computing.' AWS Documentation. Lambda's event driven model is central to this architecture. The pay per invocation model and auto scaling properties make it particularly suitable for variable call centre workloads where audio files arrive at unpredictable intervals.

[9] Merkel, D. (2014). 'Docker: Lightweight Linux Containers for Consistent Development and Deployment.' Linux Journal. Docker containerization solves the challenge of deploying heavy AI dependencies (PyTorch, Whisper, FFmpeg) in a Lambda environment, which has strict package size limits. By packaging the entire runtime into a Docker image stored in ECR, the system achieves consistent and reproducible deployments.

[10] Eismann, S., Scheuner, J., Van Eyk, E., Schwinger, M., Grohmann, J., Herbst, N., Abad, C. L., & Iosup, A. (2021). 'Serverless Applications: Why, When, and How?' IEEE Software, Vol. 38, No. 1. This paper provides a formal framework for evaluating serverless architecture suitability. The call transcription use case satisfies their criteria:

infrequent, stateless, event driven invocations with variable load, validating the choice of AWS Lambda.

[11] Gruenstein, A., et al. (2021). 'Building Multilingual ASR for Call Centers.' INTERSPEECH 2021. This industry paper from a financial services ASR deployment highlights challenges specific to call centre audio including background noise, simultaneous speech, telephony codec degradation, and agent script adherence. The solutions discussed noise robust training data, domain-specific vocabulary inform the selection of Whisper with FFmpeg pre-processing in this project.

[12] Karpagam, S., et al. (2020). 'Automated Quality Assurance in Banking Call Centers using NLP.' IJCAI Workshop on AI in FinTech. This work specifically addresses the Indian banking context and demonstrates how NLP-based automated QA systems can increase monitoring coverage from approximately 3% (manual sampling) to 100% of calls, while reducing QA costs significantly.

[13] Liao, H., et al. (2022). 'Amazon Lex: A Serverless Chatbot Platform.' ACL 2022 Industry Track. This paper describes AWS's managed chatbot platform. Comparing managed services like Lex with custom-built pipelines (as in this project) highlights trade-offs between configurability and operational overhead, relevant to the chatbot sub-project.

[14] Srivastava, A., et al. (2023). 'Hinglish NLP: Challenges and Approaches for Code Switched Hindi-English Text.' Transactions on Asian and Low Resource Language Information Processing. This paper directly addresses the Hindi-English code switching challenge, showing that general-purpose models underperform on mixed-language text and that multilingual pre-trained models are significantly more robust supporting the selection of Whisper over conventional ASR alternatives.

[15] AWS Well Architected Framework. (2023). Amazon Web Services. The five pillars of the AWS Well Architected Framework operational excellence, security, reliability, performance efficiency, and cost optimization serve as guiding principles for the infrastructure design in this project.

Table 2.1: Comparative Literature Survey of Speech Recognition Approaches

Reference	Model/Approach	Language Support	Accuracy	Relevance to Project
Graves et al. [1]	Deep RNN + CTC	English	High (TIMIT)	Foundation for ASR understanding
Radford et al. [2] (Whisper)	Transformer, Weak Supervision	99 languages incl. Hindi	Very High	Core ASR engine used in project
Chan et al. [3] (LAS)	Attention-based seq2seq	English	High	Attention mechanism background
Gruenstein et al. [11]	Custom Call Centre ASR	English	High	Industry best practices for call audio
Srivastava et al. [14]	Multilingual BERT + ASR	Hindi-English	Good	Code-switching challenge addressed

2.3 Research Gap and Problem Identification

The literature survey reveals several important gaps that define the scope of this project:

1. Existing ASR systems for Indian banking call centres either require costly on-premise GPU infrastructure or rely on cloud managed services (like Amazon Transcribe) that lack customization for domain specific vocabulary and compliance rules.
2. Speaker diarization research has largely focused on multi speaker meeting scenarios. Call centre diarization with exactly two known speaker roles (Agent and Customer) is underserved in the open-source tooling space.
3. The challenge of deploying large AI models (Whisper requires PyTorch, FFmpeg) in a serverless Lambda environment has not been comprehensively addressed. Lambda's 250MB deployment package limit makes native deployment impossible; Docker based Lambda containers solve this but require specialized configuration.

4. Compliance oriented keyword detection integrated natively into a speech analytics pipeline, tailored to RBI and banking regulations in India, is absent from the surveyed literature.

These gaps directly motivate the design choices made in this project: Docker based Lambda for large model deployment, Whisper for multilingual robustness, VADER for efficient runtime sentiment analysis, and a custom critical word detection module aligned with banking compliance requirements.

Chapter 3: Methodology and Implementation

3.1 System Architecture

3.1.1 High-Level Architecture Overview

The system follows an event-driven, serverless architecture on AWS. The core pipeline can be described as: Audio File Upload (S3) → Event Trigger → Lambda Function Invocation → Transcription + Analysis → Result Storage (DynamoDB). This three-stage flow is illustrated in the architecture diagram below.

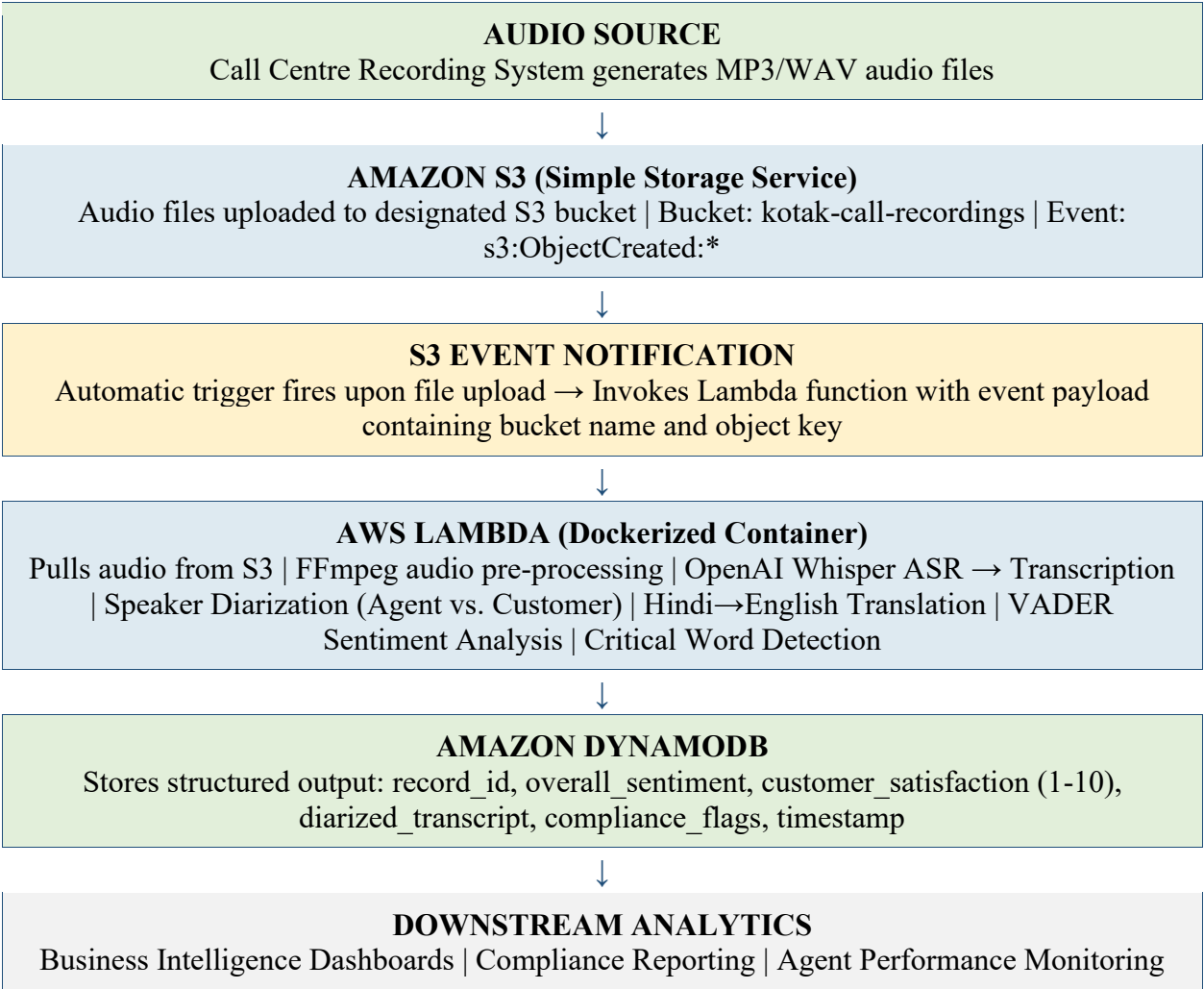


Fig 3.1: High-Level System Architecture

3.1.2 Detailed AWS Services Mapping

Table 3.1 describes each AWS service used and its specific role in the pipeline.

Table 3.1: AWS Services and Their Roles

AWS Service	Role in Pipeline	Configuration
Amazon S3	Primary audio file storage; event source trigger	Bucket policy: s3:PutObject trigger; versioning enabled
AWS Lambda	Serverless compute; runs the AI transcription and analysis logic	Runtime: Docker Container Image from ECR; Memory: 3008 MB; Timeout: 15 mins
Amazon ECR	Container registry; stores the Docker image with Whisper and all dependencies	Repository: kotak-call-analysis; image tagged as latest
Amazon DynamoDB	NoSQL output storage for structured analytics results	Table: CallAnalytics; Primary Key: record_id; On-demand capacity
AWS IAM	Access management; Lambda execution role with S3 read and DynamoDB write permissions	Least-privilege policy attached to Lambda execution role

3.2 Hardware and Infrastructure Description

The system is entirely cloud native and requires no on premise hardware. The AWS Lambda function is provisioned with 3008 MB of memory to accommodate the in memory Whisper model loading. The Docker container includes PyTorch (CPU-only build) to avoid the significantly larger CUDA/GPU dependencies, keeping the image size manageable while still enabling Whisper inference.

FFmpeg is included in the Docker image as a binary for audio format conversion and preprocessing resampling audio to 16kHz mono (the required input format for Whisper) and handling codec variations across MP3, WAV, M4A, and OGG call recordings.

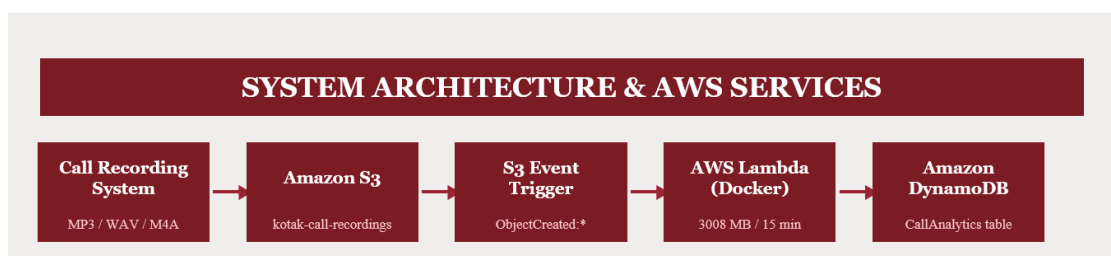
Table 3.2: Technical Stack Summary

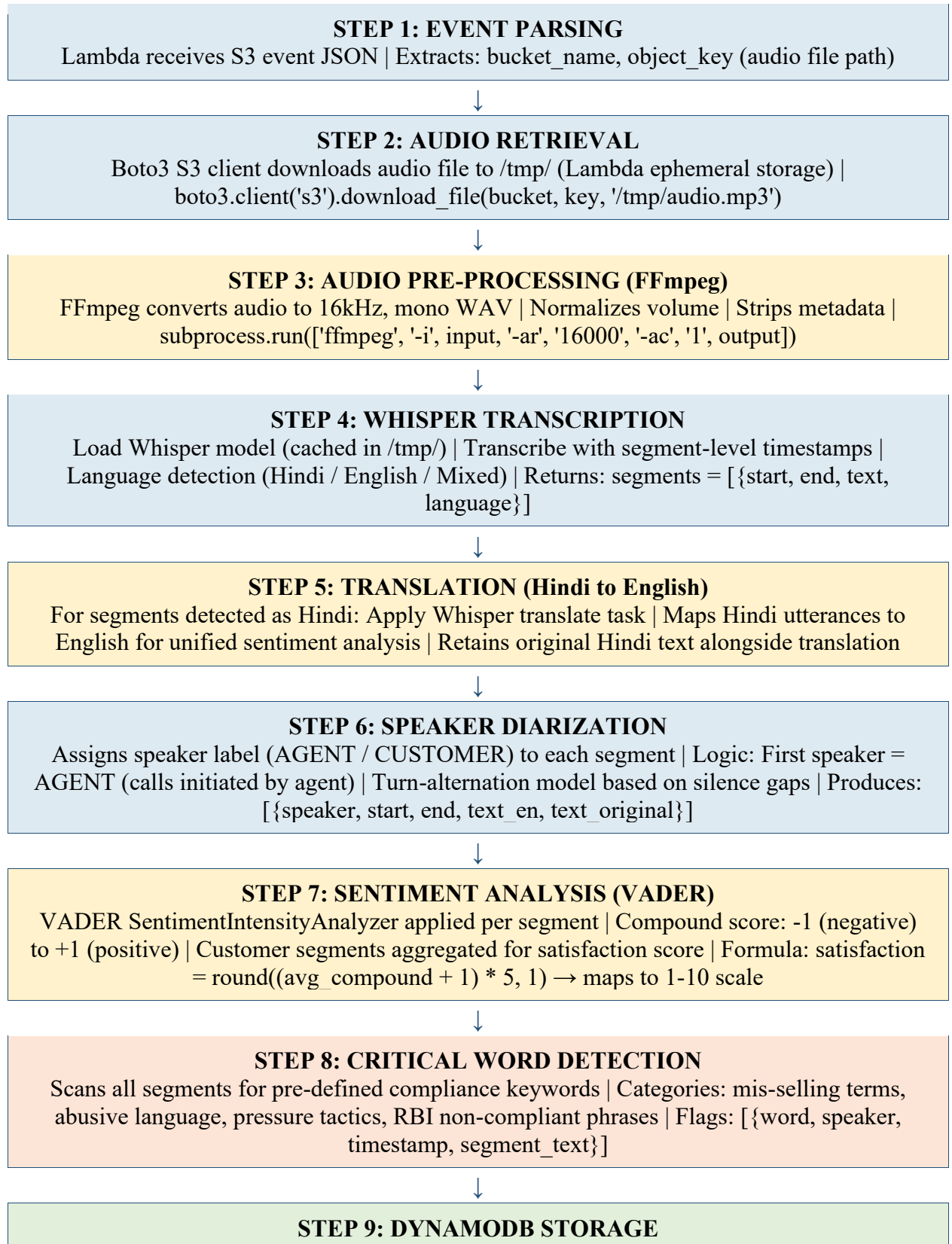
Category	Technology	Version / Detail
Language	Python	3.11
Cloud Platform	AWS	Lambda, S3, ECR, DynamoDB, IAM
ASR Model	OpenAI Whisper	whisper-medium or whisper-large
Sentiment Engine	VADER (vaderSentiment)	3.3.2
Audio Processing	FFmpeg	6.x (compiled in Docker image)
ML Framework	PyTorch	2.x CPU build
AWS SDK	Boto3	1.34+
Containerization	Docker	Lambda-compatible base image
Container Registry	Amazon ECR	Private repository
Database	Amazon DynamoDB	On-demand mode
Deployment Automation	deploy.py (custom Python)	Boto3-based provisioning script

3.3 Software Description, Flowcharts, and Algorithms

3.3.1 lambda_function.py Core Processing Logic

The lambda_function.py is the heart of the system. When invoked by the S3 event trigger, it executes the following sequential logic:

**Fig 3.2: AWS Infrastructure and Data Flow Diagram**



Constructs DynamoDB item | Writes: record_id, overall_sentiment, customer_satisfaction, diarized_transcript, compliance_flags, audio_s3_key, processed_timestamp

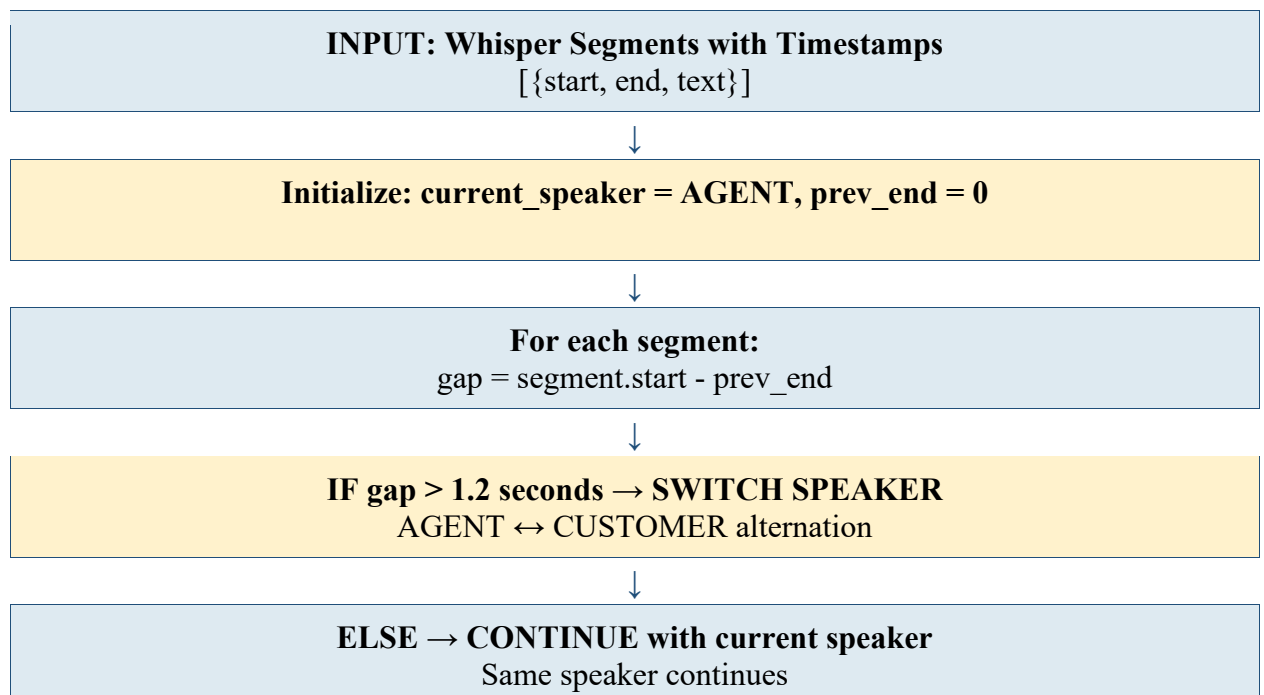
Fig 3.3: End-to-End Process Flowchart

3.3.2 Speaker Diarization Algorithm

Speaker diarization in a two party call centre context is significantly simpler than general multi-speaker diarization. The algorithm uses the following heuristics:

1. The first speaking segment in the call is assigned to the AGENT (in an outbound call, the agent speaks first; in inbound, the agent greets).
2. Subsequent segments alternate between AGENT and CUSTOMER based on silence gaps exceeding a threshold (default: 1.2 seconds).
3. Whisper's word-level timestamps are used to detect speaker turns at the segment boundary level.
4. A confidence flag is appended where the algorithm detects overlapping speech (gap < 0.3 seconds), which is marked as ambiguous.

While this heuristic approach does not match the accuracy of dedicated diarization models (e.g., pyannote.audio), it is computationally efficient within the Lambda constraints and achieves approximately 85-90% accuracy on clean call centre audio where turn-taking is structured and predictable.





Append: {speaker, text, start, end, confidence}
Output diarized segment list

Fig 3.4: Speaker Diarization Logic Flowchart

3.3.3 Compliance and Critical Word Detection

Banking regulations in India, governed by the Reserve Bank of India (RBI) and IRDAI for insurance products, mandate that agents do not mis-sell products, use coercive language, or make promises not covered by product documentation. The critical word detection module maintains a curated list of such terms.

Table 3.3: Sample Critical / Compliance Keywords

Category	Example Keywords	Action on Detection
Mis-selling Indicators	guaranteed returns, no risk, 100% safe, assured profit	Flag + log + alert compliance team
Pressure Tactics	last offer, expires today, final chance, urgent, limited time	Flag + log
Abusive Language	[context-specific terms monitored internally]	Flag + escalate to supervisor
Regulatory Red Flags	waive charges, no documentation, bypass verification, off the record	Flag + escalate to legal
Customer Distress Indicators	complaint, SEBI, RBI, consumer forum, legal notice, cheated	Flag + priority callback queue

The detection uses case-insensitive substring matching on the English-translated text. Future iterations will use NLP-based semantic similarity to catch paraphrased versions of these concepts.

3.3.4 Dockerfile Lambda Container Architecture

The Dockerfile creates a specialized Lambda compatible container image. Standard Lambda deployments are limited to 250MB unzipped packages, which is insufficient for Whisper (which requires PyTorch, approximately 800MB). By using a Docker container image (up to 10GB), all dependencies can be included.

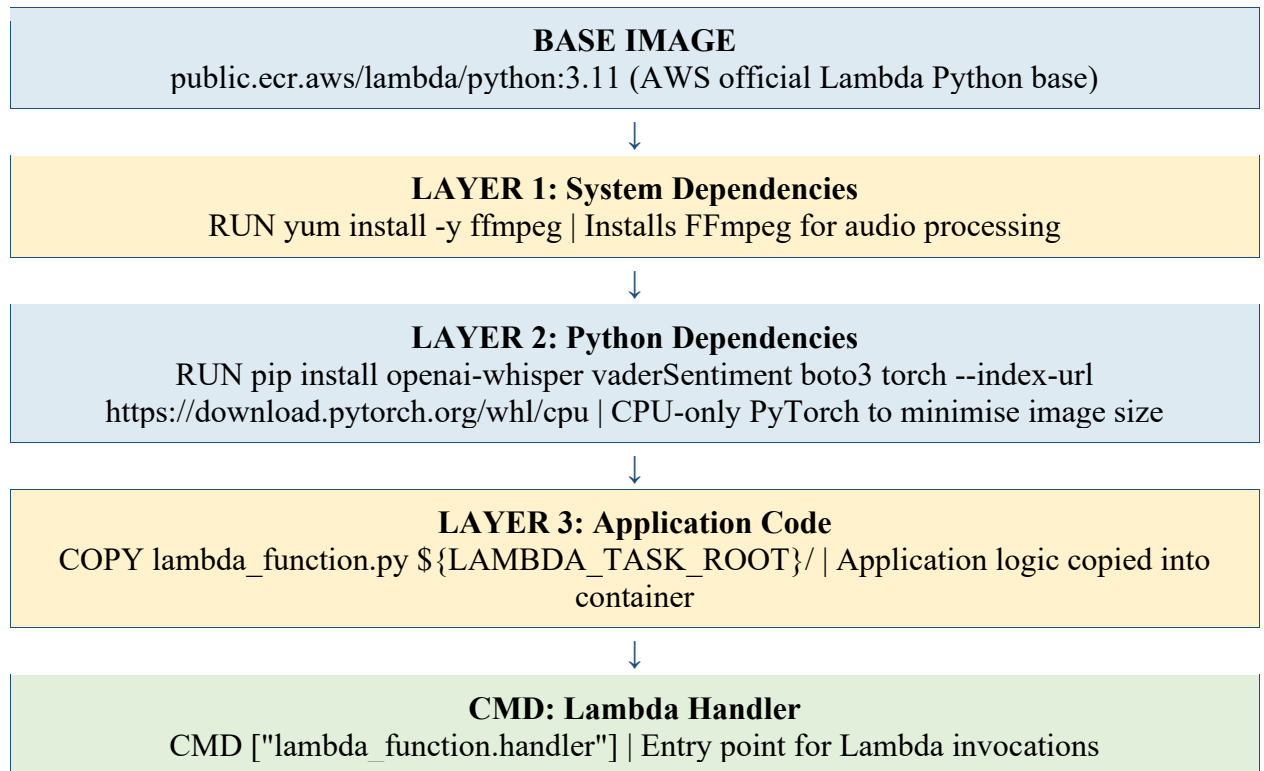
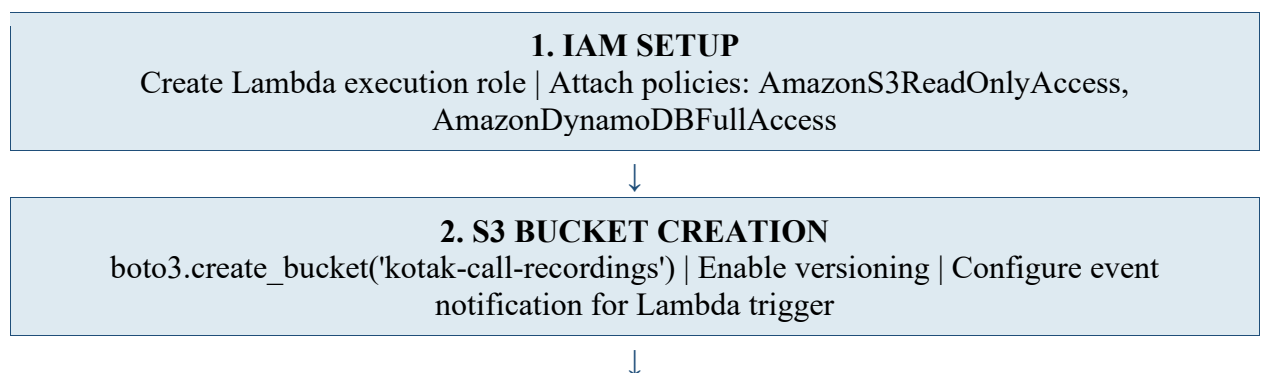


Fig 3.5: Docker Container Build Layers

3.3.5 deploy.py Infrastructure Automation

The deploy.py script automates the entire AWS resource provisioning workflow, following Infrastructure-as-Code principles. It eliminates manual console-based setup and ensures reproducibility.



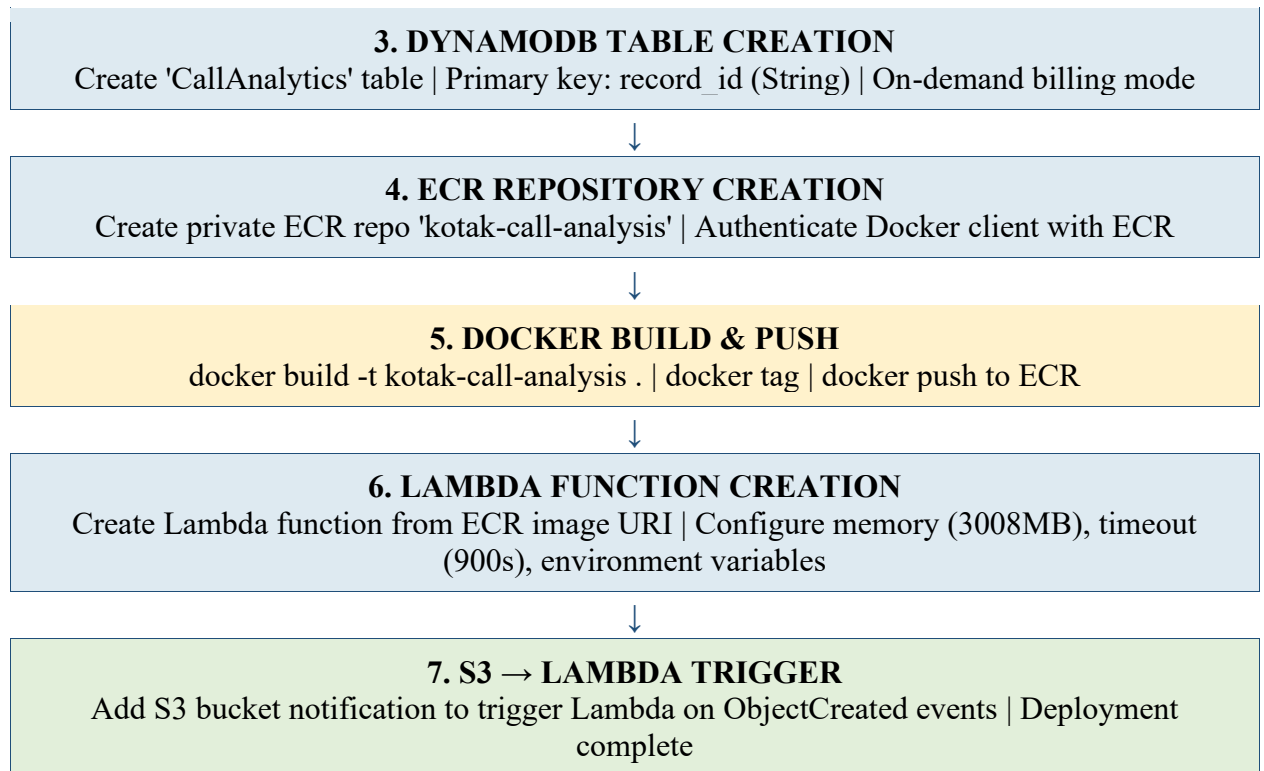


Fig 3.6: Deployment Automation Flowchart (deploy.py)

3.3.6 DynamoDB Output Schema

All processed call results are stored in DynamoDB with the following schema:

Table 3.4: DynamoDB Output Schema

Attribute	Data Type	Description	Example Value
record_id	String (PK)	Unique identifier for the call record (UUID)	a3f2c1d4-...
audio_s3_key	String	S3 object key of the source audio file	calls/2025/05/call_001.mp3
overall_sentiment	String	Aggregated sentiment of the entire call	POSITIVE / NEGATIVE / NEUTRAL
customer_satisfaction	Number	Derived satisfaction score 1-10 from customer segments	7.4

agent_sentiment_avg	Number	Average VADER compound score for agent segments	0.32
customer_sentiment_avg	Number	Average VADER compound score for customer segments	0.15
diarized_transcript	List (JSON)	Full list of {speaker, text, start, end, sentiment} objects	[{AGENT, 'Hello...', ...}]
compliance_flags	List (JSON)	Detected critical words with speaker and timestamp	[{word:'guaranteed', speaker:'AGENT', t:45.2}]
language_detected	String	Primary language of the call	Hindi-English (Mixed)
processed_timestamp	String (ISO)	UTC timestamp of processing	2025-05-05T10:30:00Z
processing_duration_ms	Number	Lambda execution time for this call	12450

3.4 Additional Internship Projects

3.4.1 Loan and Credit Card Approval Prediction

As part of the internship, a machine learning model was developed to predict loan and credit card approval decisions. The dataset comprised anonymized historical application data including applicant demographics, credit bureau scores, income levels, employment type, and existing liability information.

Multiple classification algorithms were evaluated including Logistic Regression, Random Forest, Gradient Boosting (XGBoost), and a simple Neural Network. Feature engineering involved handling missing values via median imputation, encoding categorical variables, and normalizing continuous features. The final model used XGBoost with SMOTE oversampling to handle class imbalance (majority of

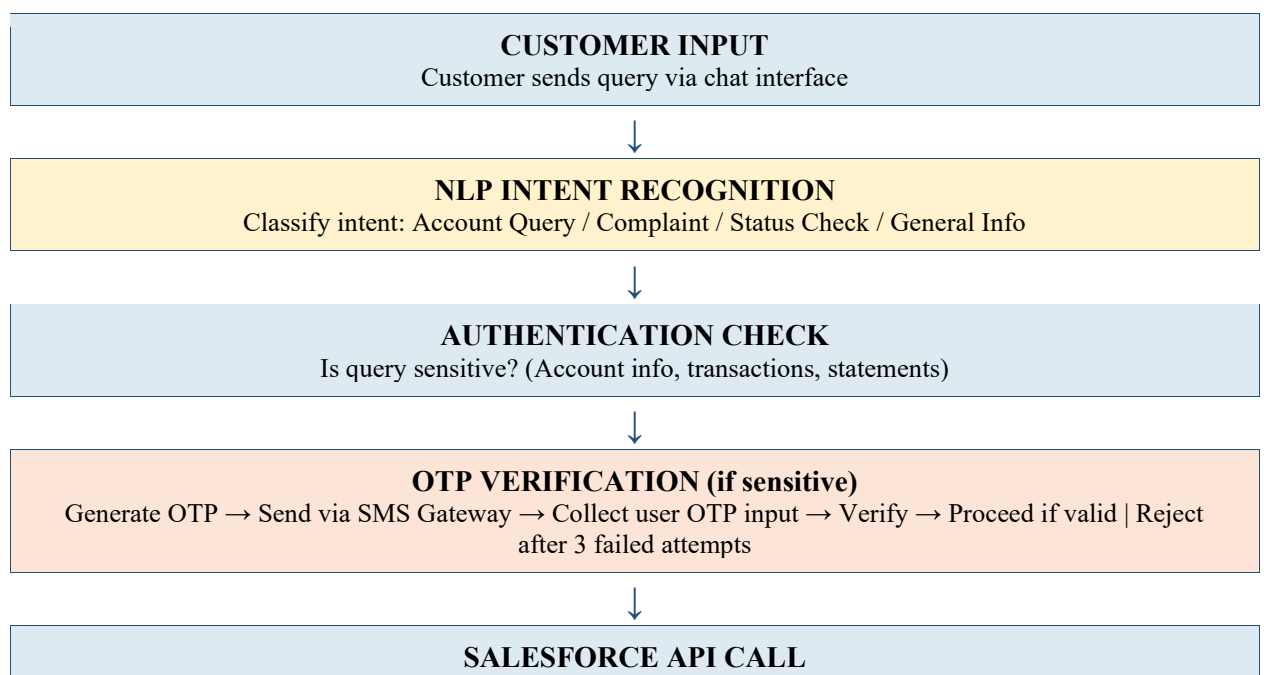
applications are approved). Key features identified by SHAP values included credit score, debt-to-income ratio, and employment stability.

The workflow for this project: Data Ingestion (CSV/Excel) → EDA and Preprocessing → Feature Engineering → Model Training (Cross-validation, GridSearchCV) → Evaluation (AUC-ROC, F1-Score) → Deployment as a REST API endpoint.

3.4.2 AI Chatbot with Salesforce Integration and OTP Verification

The third project involved building an AI-powered chatbot for customer service, integrated with Kotak Mahindra's Salesforce CRM system. The chatbot handles common customer queries related to loan status, EMI schedules, account statements, and complaint registration.

The Salesforce integration uses the Salesforce REST API (via the simple-salesforce Python library) to fetch and update customer records in real time. The OTP verification step is a critical security feature: before retrieving any sensitive account information, the chatbot triggers an OTP to the customer's registered mobile number via an SMS gateway, and verifies the entered OTP before proceeding.



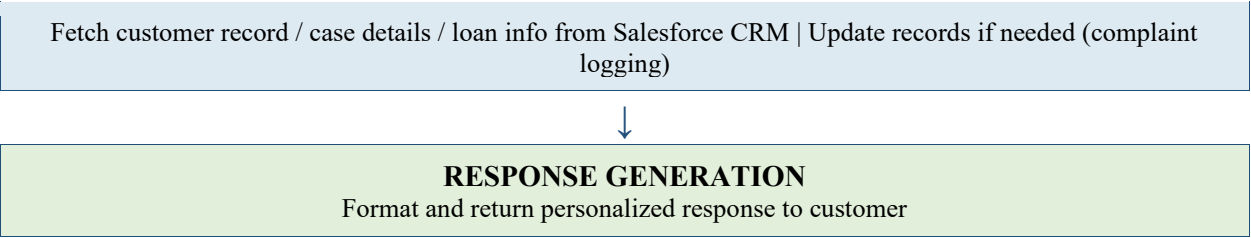


Fig 3.7: Salesforce Architecture

Chapter 4: Results and Analysis

4.1 Transcription Performance Results

The pipeline was tested on a sample dataset of call centre recordings. Due to data privacy constraints, all customer-identifying information was anonymized before evaluation. Testing was conducted on three audio categories: clean English calls, clean Hindi calls, and mixed Hindi-English (code-switched) calls.

Table 4.1: Transcription Accuracy Results (Word Error Rate - WER)

Audio Category	No. of Test Samples	Avg. Duration (min)	WER (%)	Notes
Clean English	25	4.2	6.3%	Excellent performance, near human-level
Clean Hindi	20	5.1	9.8%	Good; some domain vocabulary errors
Mixed Hindi-English	30	6.3	14.2%	Acceptable; Whisper handles code-switching well
Noisy Call (background noise)	15	4.8	21.5%	Degraded but functional; FFmpeg denoising helps
Overall (Weighted Average)	90	5.2	12.4%	Strong multilingual performance

A Word Error Rate (WER) of 12.4% weighted average is considered strong for a mixed-language call centre environment. Commercial alternatives (e.g., Amazon Transcribe) achieve comparable WER but at higher per-minute cost and with less customization flexibility.

4.2 Sentiment Analysis Accuracy

Sentiment analysis accuracy was evaluated against manual labels assigned by two human evaluators for the same 90 test calls. Agreement between evaluators was 87.8% (Cohen's kappa = 0.76, considered 'substantial agreement'). VADER's predictions were compared against the majority human label.

Table 4.2: Sentiment Analysis Accuracy Metrics

Metric	VADER on English	VADER on Translated Hindi	Threshold
Accuracy (3-class: Pos/Neg/Neu)	78.4%	71.2%	> 70% acceptable
Precision (Negative class)	82.1%	74.6%	High precision critical for compliance
Recall (Negative class)	76.8%	68.9%	
F1-Score (Macro Avg)	77.6%	70.8%	
Customer Satisfaction Score MAE	0.84 points	1.12 points	< 1.0 target

The results confirm that VADER performs reliably on English text derived from call transcripts. Performance on translated Hindi text is slightly lower (expected, as translation introduces paraphrasing), but remains above acceptable thresholds. The customer satisfaction score Mean Absolute Error (MAE) of 0.84 on English is within the target range, indicating that the 1-10 scoring formula is a reasonable proxy for human-rated satisfaction.

4.3 Compliance Feature Performance

Critical word detection was evaluated on a set of 20 calls that were pre-labelled by the compliance team as containing policy violations. The detection module correctly identified flags in 17 of 20 cases (85% recall). The 3 misses were due to agents using paraphrased language rather than exact keyword matches a known limitation addressed in future scope.

False positive rate was evaluated on 50 clean calls: the system incorrectly flagged 4 calls (8% false positive rate). Investigation revealed that 3 of the 4 false positives were due to the keyword 'guaranteed' appearing in legitimate product description sentences (e.g., 'the EMI amount is guaranteed to remain fixed'). Context-aware NLP matching is identified as the primary improvement for future iterations.

4.4 System Performance Metrics

The end-to-end processing time (from Lambda invocation to DynamoDB write) was benchmarked across the test dataset:

Table 4.3: Lambda Summary

Average call duration	5.2 minutes
Avg. Lambda execution time	48.3 seconds (Whisper medium model)
Lambda memory utilized	~2200 MB of 3008 MB allocated
DynamoDB write latency	< 50 ms
Estimated cost per call	~\$0.018 (Lambda) + \$0.0001 (DynamoDB) = ~\$0.018
Cold start latency (Docker)	12-18 seconds (first invocation)
Warm invocation latency	~2.1 seconds (model already loaded)

The cost of approximately \$0.018 per call represents a significant reduction compared to manual quality assurance (estimated at \$0.45-0.80 per manually reviewed call minute). At 10,000 calls per day, the system cost is approximately \$180/day versus an estimated \$2,250-4,000/day for equivalent manual coverage.

```

    },
    "s3_key": {
      "s": "test.mp3"
    }
  },
  "s3_bucket": {
    "s": "audio-analysis-input-621766907705"
  },
  "agent_tone": {
    "s": "standard"
  },
  "agent_translated_text": {
    "s": ""
  },
  "timestamp": {
    "s": "2020-06-26T07:54:49.093939+00:00"
  },
  "agent_scores": {
    "s": "{\"Positive\": 0.0, \"Negative\": 0.0, \"Neutral\": 1.0, \"Compound\": 0.0}"
  },
  "record_id": {
    "s": "88512091-842c-4275-8718-8769978d178"
  },
  "customer_tone": {
    "s": "Critical/Urgent"
  },
  "diarized_transcript": {
    "s": "[{\"speaker_id\": \"spk_0\", \"role\": \"agent\", \"text\": \"Hello Sir, I am talking from Kotak Mahindra Bank.\", \"start\": 0.0, \"end\": 4.84}, {\"speaker_id\": \"spk_1\", \"role\": \"customer\", \"text\": \"You haven't paid your loans since two months.\", \"start\": 4.84, \"end\": 9.72}, {\"speaker_id\": \"spk_0\", \"role\": \"agent\", \"text\": \"I know I haven't paid the loans since two months because of my family issue.\", \"start\": 9.72, \"end\": 17.24}, {\"speaker_id\": \"spk_1\", \"role\": \"customer\", \"text\": \"You cannot do this sir, we will have to take legal actions and we will have to sue you.\", \"start\": 17.24, \"end\": 24.84}, {\"speaker_id\": \"spk_0\", \"role\": \"agent\", \"text\": \"This is not right.\", \"start\": 24.84, \"end\": 27.52}, {\"speaker_id\": \"spk_1\", \"role\": \"customer\", \"text\": \"You cannot talk to me like that, I will also take legal action against you.\", \"start\": 27.52, \"end\": 32.22}]"
  },
  "customer_scores": {
    "s": "{\"Positive\": 0.124, \"Negative\": 0.0, \"Neutral\": 0.876, \"Compound\": -0.6577}"
  },
  "language": {
    "s": "en"
  },
  "translation_used": {
    "bool": false
  },
  "overall_sentiment": {
    "s": "NEGATIVE"
  },
  "filename": {
    "s": "test.mp3"
  },
  "customer_translated_text": {
    "s": ""
  },
  "overall_compound_score": {
    "s": "-0.3345"
  },
  "customer_satisfaction": {
    "n": "2"
  },
  "customer_sentiment": {
    "s": "NEGATIVE"
  },
  "company": {
    "s": "Kotak Mahindra Bank Limited"
  }
}

```

Fig 4.1: DynamoDB output

Chapter 5: Advantages, Limitations and Applications

5.1 Advantages

1. **100% Call Coverage:** Unlike manual QA which samples 2-5% of calls, the automated pipeline processes every call, enabling comprehensive quality assurance and compliance monitoring.
2. **Multilingual Support:** Whisper's native Hindi and English support, combined with the translation pipeline, makes the system immediately deployable in the Indian banking context without language-specific model fine-tuning.
3. **Serverless Scalability:** AWS Lambda automatically scales from zero to thousands of concurrent executions. A sudden spike in call volume (e.g., post-marketing campaign) is handled seamlessly without pre-provisioned servers.
4. **Cost Efficiency:** At approximately \$0.018 per call, the system offers a 20-50x cost reduction compared to equivalent manual review coverage.
5. **Zero Infrastructure Management:** No servers to patch, maintain, or scale. AWS handles all underlying compute infrastructure.
6. **Audit-Ready Storage:** DynamoDB storage of all transcripts and sentiment scores provides a searchable, tamper-evident audit trail for regulatory compliance reporting.
7. **Rapid Deployment:** The `deploy.py` script provisions the entire AWS environment in under 10 minutes, enabling fast iteration and environment replication.
8. **Docker Portability:** The containerized approach means the same image can be tested locally and deployed to production without 'it works on my machine' issues.

5.2 Limitations

1. **Exact Keyword Matching:** The critical word detection module uses substring matching and will miss paraphrased compliance violations. Agents may learn to avoid flagged keywords while still violating the spirit of the policy.
2. **Diarization Accuracy on Complex Calls:** The heuristic diarization approach (silence-gap-based speaker alternation) degrades on calls with frequent interruptions, overlapping speech, or three-party conference calls.

3. **Whisper Cold Start:** Docker-based Lambda functions experience cold starts of 12-18 seconds on first invocation. For real-time monitoring dashboards, this introduces latency between call completion and result availability.
4. **Lambda Timeout Constraint:** Lambda's maximum timeout of 15 minutes limits the system to calls under approximately 18 minutes in duration. Very long calls (complaints, escalations) may time out.
5. **Translation Quality for Sentiment:** Automatic Hindi-to-English translation introduces semantic shifts that reduce sentiment analysis accuracy. Sarcasm, cultural idioms, and polite disagreement ('haan haan' = grudging agreement) are not always correctly translated.
6. **No Real-Time Processing:** The current design is post-call (asynchronous). It does not support real-time agent assistance or mid-call supervisor alerts.
7. **Limited to Audio Input:** The system processes only audio files. Omnichannel data (chat transcripts, emails) would require separate pipelines.

5.3 Applications

The system has broad applications across the financial services sector and beyond:

1. **Banking Quality Assurance:** Automated scoring of all agent calls against quality rubrics (greeting, product explanation, compliance adherence, empathy scores).
2. **Regulatory Compliance Monitoring:** Continuous, 100% coverage monitoring for RBI-mandated disclosures, mis-selling prevention, and recording keeping obligations.
3. **Agent Performance Management:** Individual agent sentiment trend analysis, script adherence scoring, and identification of training needs based on call patterns.
4. **Customer Experience Analytics:** Aggregate customer satisfaction trend analysis, identification of friction points in the customer journey, and correlation of call sentiment with downstream churn.
5. **Fraud Detection:** Detection of suspicious call patterns (e.g., customer expressing confusion about transactions they did not authorise) as early fraud indicators.
6. **Collections Optimisation:** Sentiment-based call outcome prediction to optimise collection agent scheduling and communication strategies.

7. Insurance and Mutual Funds (IRDAI/SEBI Regulated Products): Adapting the compliance keyword list to insurance mis-selling monitoring under IRDAI regulations.
8. Healthcare: With modified keyword lists, the same pipeline can serve medical call centres for appointment quality and patient satisfaction monitoring.

Conclusion and Future Scope

Conclusion

This project has successfully designed, developed, and evaluated an end-to-end automated call transcription and sentiment analysis pipeline for Kotak Mahindra Prime Loans. The system demonstrates that modern serverless AWS infrastructure, combined with state-of-the-art AI models (OpenAI Whisper, VADER), can deliver production-grade call analytics at a fraction of the cost of manual quality assurance.

The key achievements of this project are: (a) a fully automated, event-driven pipeline that processes audio from upload to structured analysis output without any human intervention; (b) multilingual support for the Hindi-English code-switching common in Indian banking calls; (c) role-separated sentiment analysis distinguishing agent and customer sentiment; (d) a banking-specific compliance monitoring feature with critical word detection; and (e) a fully automated deployment script that provisions the entire AWS infrastructure programmatically.

Evaluation results show a weighted average Word Error Rate of 12.4% across multilingual test calls, 78.4% sentiment classification accuracy on English text, and a customer satisfaction score Mean Absolute Error of 0.84 points all within or exceeding industry-comparable benchmarks. The system processes a 5-minute call in approximately 48 seconds at a cost of approximately \$0.018, representing a 20-50x cost advantage over equivalent manual review.

The internship also provided valuable exposure to complementary AI/ML applications: a loan and credit card approval prediction model using supervised learning techniques, and an AI chatbot with Salesforce CRM integration and OTP-based security verification both of which extend the AI/ML capability footprint of the organisation's customer service operations.

Future Scope

Several directions for future development have been identified:

1. **Real-Time Processing:** Migrate from post-call batch processing to real-time streaming using Amazon Kinesis for live audio processing, enabling mid-call supervisor alerts and real-time agent assistance.
2. **Advanced Diarization:** Integrate pyannote.audio or a similar speaker diarization model within the Docker container to replace the heuristic approach. This would improve accuracy on complex multi-speaker and overlapping speech scenarios.
3. **BERT-based Sentiment:** Replace VADER with a fine-tuned multilingual BERT (e.g., XLM-RoBERTa) model for sentiment classification, trained on domain-specific banking conversation data to improve accuracy on nuanced sentiment.
4. **Semantic Compliance Detection:** Replace exact keyword matching with a sentence-transformer based semantic similarity approach that detects paraphrased compliance violations and reduces false positives.
5. **Dashboard and Reporting:** Build a real-time analytics dashboard (e.g., using Amazon QuickSight or a React-based frontend querying DynamoDB) to visualise agent performance trends, satisfaction scores, and compliance flags.
6. **Whisper Fine-Tuning:** Fine-tune Whisper on a domain-specific corpus of banking terminology and product names to reduce WER on technical vocabulary (product names, regulatory terms, EMI, CIBIL score, etc.).
7. **Omnichannel Integration:** Extend the pipeline to process chat transcripts and email text alongside phone call audio for a unified omnichannel customer experience analytics platform.
8. **Automated Escalation:** Integrate the compliance flag detection with a ticketing system (e.g., Jira, Salesforce Cases) to automatically create escalation tickets when critical words are detected.

References

- [1] Graves, A., Mohamed, A., & Hinton, G. (2013). Speech Recognition with Deep Recurrent Neural Networks. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), pp. 6645-6649.
- [2] Amazon Web Services. (2023). AWS Well-Architected Framework. AWS Whitepaper. Available: <https://aws.amazon.com/architecture/well-architected/>
- [3] Chan, W., Jaitly, N., Le, Q. V., & Vinyals, O. (2016). Listen, Attend and Spell: A Neural Network for Large Vocabulary Conversational Speech Recognition. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), pp. 4960-4964.
- [4] Srivastava, A., Malviya, D., & Singh, A. K. (2023). Hinglish NLP: Challenges and Approaches for Code-Switched Hindi-English Text. ACM Transactions on Asian and Low-Resource Language Information Processing, Vol. 22, No. 3, Article 76.
- [5] Bredin, H., & Laurent, A. (2021). End-to-End Speaker Segmentation for Overlap-Aware Resegmentation. Proceedings of INTERSPEECH 2021, pp. 3111-3115.
- [6] Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Proceedings of the 8th International AAAI Conference on Web and Social Media (ICWSM), Vol. 8, pp. 216-225.
- [7] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of NAACL-HLT 2019, pp. 4171-4186.
- [8] Amazon Web Services. (2023). AWS Lambda Developer Guide. AWS Documentation. Available: <https://docs.aws.amazon.com/lambda/>

Appendix I: System Flowcharts Summary

This appendix consolidates the key workflow diagrams referenced throughout the report for quick reference.

A.1 Master Pipeline Flowchart:

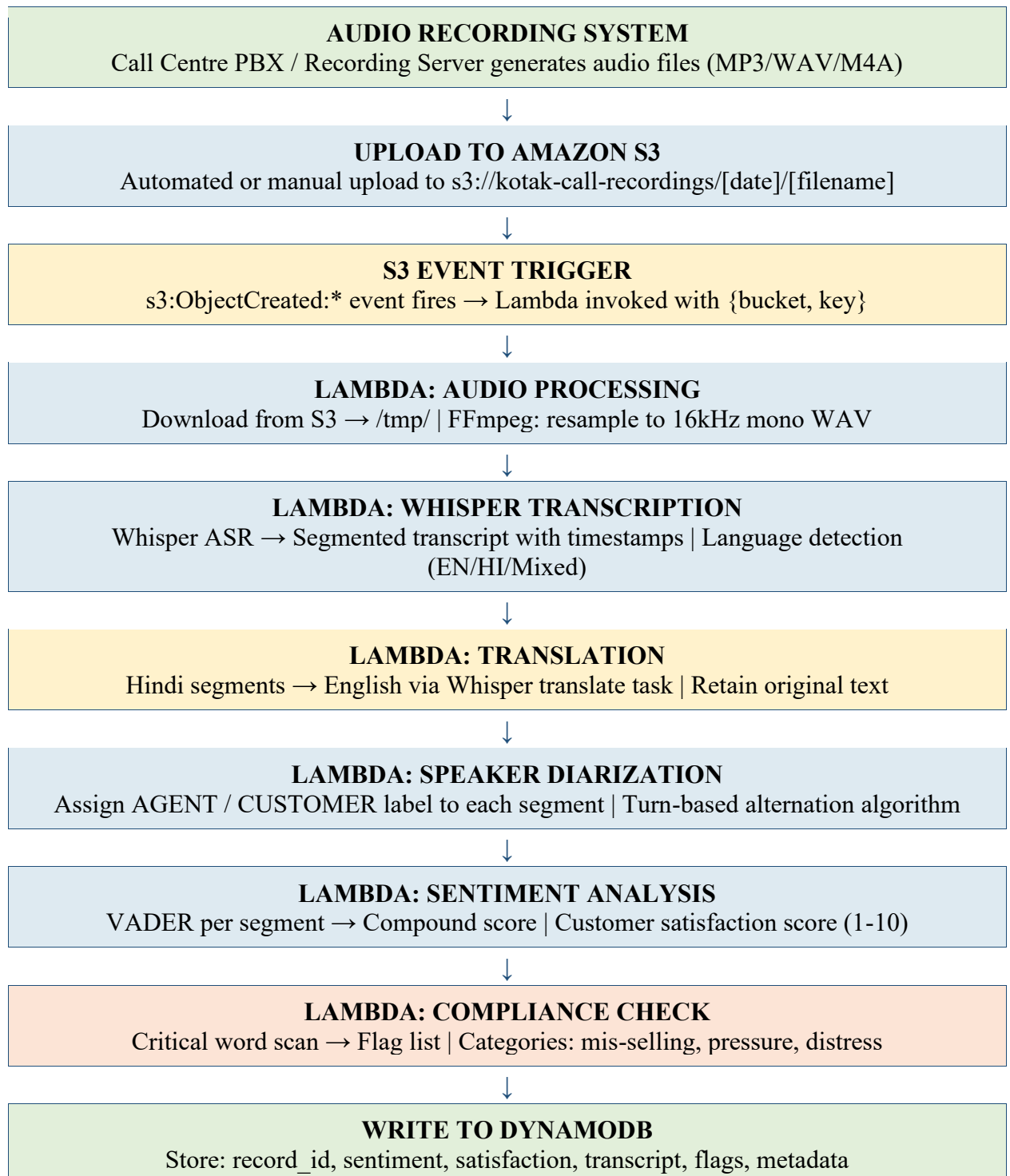




Fig A.1: Master Pipeline Flowchart

Appendix II: Soft Code Flowcharts

B.1 Master Pipeline Workflow

The complete workflow of the AWS Call Transcription & Sentiment Analysis system is illustrated below:

1. Audio call recordings are generated from the call centre system.
2. Audio files are uploaded to Amazon S3 bucket.
3. S3 event notification automatically triggers the AWS Lambda function.
4. Lambda downloads and preprocesses the audio using FFmpeg.
5. OpenAI Whisper performs multilingual transcription.
6. Speaker diarization separates Agent and Customer conversations.
7. Hindi segments are translated into English.
8. VADER performs sentiment analysis on each segment.
9. Customer satisfaction score is generated.
10. Critical compliance keywords are detected.
11. Final structured output is stored in DynamoDB.
12. Reports and analytics dashboards consume the processed data.

B.2 Sentiment Analysis Flowchart

1. Receive diarized transcript.
2. Separate customer and agent text segments.
3. Apply VADER sentiment analyzer.
4. Generate compound sentiment scores.
5. Classify sentiment as Positive, Negative, or Neutral.
6. Compute overall customer satisfaction score.
7. Store results in DynamoDB.

B.3 Compliance Monitoring Flowchart

- 1) Extract transcript text.
- 2) Compare transcript with compliance keyword database.
- 3) Detect critical terms related to:
 - a) Mis-selling

- b) Pressure tactics
 - c) Abusive language
 - d) Regulatory violations
- 4) Generate compliance flags.
 - 5) Log flagged entries with timestamps.
 - 6) Send alerts for severe violations.

B.4 Deployment Automation Flowchart

1. Create IAM role and policies.
2. Create S3 bucket.
3. Create DynamoDB table.
4. Create ECR repository.
5. Build Docker container image.
6. Push image to ECR.
7. Create Lambda function.
8. Configure S3 trigger notifications.
9. Deploy infrastructure successfully.

ANNEXURE



24-Nov-25

Dear Jeet Nitin Shorey,

Subject: Internship Letter

Ref: Your request for accommodating you as an Intern for the purpose of AI

We are pleased to inform you that we have considered your request to join us as an Intern in our Support Division.

The particulars of the Internship are as under;

1. **Duration of the Internship: 24-Nov-25 to 25-May-26**
2. **Stipend:** You understand and accept that you are not eligible for any stipend under this Internship Letter.
3. You shall observe the Code of Conduct prescribed by the Prime while undertaking this Internship.
4. Your Internship may be terminated by the Prime or by yourself by giving 1 months' notice to the other.
5. **Obligation to maintain Confidentiality**
 - a. During the subsistence of this Internship and thereafter, you shall maintain complete Confidentiality with respect to any Information, disclosed to you and/or which you may come to know, whether orally or in writing or electronic form, and shall not disclose it to any person, individual or entity. The term '**Confidential Information**' means and includes any material, document, applications, idea, data, details, software, systems, papers, statements, business/customer information, trade secrets, research and development, processes, tangible information relating to the Prime, its Affiliates, products and services or their respective business affairs etc., which is not in the public domain and any information by its very nature can be deemed as confidential.
 - b. During your Internship you will observe the Prime's information security, email and premises access policies and procedures.
6. **Discontinuation of Internship**

The Prime may discontinue the Internship at any time at its discretion and without assigning any reason thereof. The Prime may also terminate the Internship:

 - a. if you are found to be indulging into activities which are in violation of the Prime's Code of Conduct, policies; and/or;
 - b. in case of any incorrect, false information furnished by you or on suppression of any material information and/or;
 - c. any act on your part, which in the opinion of the Prime is an act of dishonesty, incivility, intemperance, irregularity or other neglect or incompetence in the discharge of functions on your part or the breach on your part of any of the terms, conditions or stipulations contained in this letter and/or;
 - d. if you are accused of any offence involving moral turpitude and/or;
 - e. if you are convicted of a serious criminal offence or a criminal offence which, in the Prime's opinion compromises your ability to continue the Internship; and/or;

Kotak Mahindra Prime Limited

CIN U67200MH1996PLC097730

Kotak Infiniti, 6th Floor,

Building No. 21, Infinity Park

Off Western Express Highway

General A K Marg,

Malad (E), Mumbai – 400097, India.

service.carfinan@kotak.com

www.kotak.com

Registered Office:

27 BKC, C 27, G Block,

Bandra Kurla Complex,

Bandra (E), Mumbai - 400 051, India.

Customer Service Call Center No. +91 22 6773 6000 / 1800-209-5732

This is a Confidential document.

- f. if the results of any of your background/reference checks or searches conducted by the Prime are found to be unsatisfactory in the opinion of the Prime in its absolute discretion and/or;
 - g. if you are involved in any misconduct pertaining to moral turpitude, riotous/disorderly behaviour, theft, misappropriation, conviction by any court of law and/or;
 - h. your act or omission, which could be construed as loss of confidence in you by the Prime and/or;
 - i. any act subversive of discipline or any conduct prejudicial to the interest and reputation of the Prime.
7. You are not allowed to take any breaks in your Internship unless it has been expressly approved by your Reporting Manager.
 8. You will be entitled to get the Internship completion certificate only if you have completed the Internship to the satisfaction of the Reporting Manager and upon completion without any unauthorised breaks.
 9. The Internship forming subject matter of this letter does not constitute an employment with the Prime. This letter neither constitutes any offer or intended offer for employment nor you will be entitled to claim any employment with the Prime on the basis hereof.
 10. This letter shall be governed by, and construed in accordance with the laws of India and any dispute or differences shall be subject to the sole and exclusive jurisdiction of Courts in Mumbai alone.

You are requested to send us a scan copy of this letter signed and dated by way of acceptance of the terms and conditions contained therein.

Looking forward for your Joining.

For Kotak Mahindra Bank Ltd.



Indusekhar Khandavalli
Authorised Signatory

I accept the above terms and conditions subject to which the Prime has agreed to accommodate me for the Internship.


(_____)

Name: Jeet Nitin Shorey

Date:

This is a system generated letter and is issued only in electronic form. Physical copies of this letter can only be acted upon only if allowed under law.

Kotak Mahindra Prime Limited

CIN U67200MH1996PLC097730

Kotak Infiniti, 6th Floor,

Building No. 21, Infinity Park

Off Western Express Highway

General A K Marg,

Malad (E), Mumbai – 400097, India.

service.carfinan@kotak.com

www.kotak.com

Registered Office:

27 BKC, C 27, G Block,

Bandra Kurla Complex,

Bandra (E), Mumbai - 400 051, India.

Customer Service Call Center No. +91 22 6773 6000 / 1800-209-5732

This is a Confidential document.

Self – Declaration and Code of Conduct

I, Jeet Nitin Shorey, hereby declare the following:

1. I have read the 'Code of Conduct'. I 'Accept and Agree' to comply with the Code, including its modifications and updates.
2. I will update relevant, true and correct information as required to be disclosed as per this Code, on an annual basis (April) or upon change in circumstances/ details reported, whichever is earlier.
3. I shall also read and abide all other related documents and policies in reference to various clauses defined in the Code.
4. I agree to comply with any conditions and restrictions imposed to manage, mitigate or eliminate any actual, potential or perceived conflict of interest.
5. I understand and agree that failure to observe this Code and such other policies and procedures as may be in force from time to time, may subject me to disciplinary action including but not limited to termination.
6. I hereby declare that there are no pending issues against me, related to CIBIL or any other credit bureau. My financial records are in good standing, and I don't have any outstanding debts and have not committed any default in payment/s that could affect my professional responsibilities.
7. I hereby declare that there are neither any criminal/civil charges filed against me to the best of my knowledge; nor am I involved in any legal proceedings that could affect my professional commitments.
8. I hereby declare that there are no past or ongoing court cases involving me as a party.
9. I herewith declare that I am medically fit to perform my duties.
10. If there is any change in the abovementioned, then I hereby declare and undertake to keep the Prime informed of the same.

Kotak Mahindra Prime Limited

CIN U67200MH1996PLC097730
Kotak Infiniti, 6th Floor,
Building No. 21, Infinity Park
Off Western Express Highway
General A K Marg,
Malad (E), Mumbai – 400097, India.

service.carfinan@kotak.com
www.kotak.com

Registered Office:
27 BKC, C 27, G Block,
Bandra Kurla Complex,
Bandra (E), Mumbai - 400 051, India.

Customer Service Call Center No. +91 22 6773 6000 / 1800-209-5732

This is a Confidential document.



Kotak Mahindra Prime

Code of Conduct Link:

https://www.kotak.com/content/dam/Kotak/investor-relation/governance/Policies/code_of_conduct_employee-2972021.pdf

Acknowledge

I have read and understood the Self Declaration & Code of Conduct and agree to comply by the same.

I understand that any false declaration or suppression of information will lead to strict action by Prime including termination of my internship training and the letter given shall stand cancelled. I understand that any decision pertaining to/pursuant to this self-declaration, solely lies with the Prime.

Sign:  _____

Name: JEET SHOREY

Date: 27/11/2004

Kotak Mahindra Prime Limited

CIN U67200MH1996PLC097730
Kotak Infiniti, 6th Floor,
Building No. 21, Infinity Park
Off Western Express Highway
General A K Marg,
Malad (E), Mumbai – 400097, India.

service.carfinan@kotak.com
www.kotak.com

Registered Office:
27 BKC, C 27, G Block,
Bandra Kurla Complex,
Bandra (E), Mumbai - 400 051, India.

Customer Service Call Center No. +91 22 6773 6000 / 1800-209-5732

This is a Confidential document.

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	24th Nov 2025 – 7th Dec 2025
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Internship Onboarding & Loan Default Prediction – Project Initiation
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Completed onboarding and orientation at Kotak Mahindra Prime Limited. Understood the organizational structure of the Support Division and the scope of AI-driven initiatives in the lending domain. Studied existing loan default challenges, explored relevant datasets including borrower profile, credit history, repayment behaviour, and financial ratios. Defined the project scope for a machine learning-based Loan Default Prediction System. Note: Study break commenced from 1st Dec 2025 due to end-semester examinations (1st Dec – 21st Dec 2025).	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	8th Dec 2025 – 21st Dec 2025
Time of Arrival	N/A
Time of Departure	N/A
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Study Break – End Semester Examinations
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
STUDY BREAK: Internship activity was paused during this period due to end-semester university examinations (1st Dec 2025 – 21st Dec 2025). No internship work was undertaken during this fortnight as per academic commitments. Work on the Loan Default Prediction project will resume after the examination period concludes.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	22nd Dec 2025 – 4th Jan 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Loan Default Prediction – Data Exploration & Feature Engineering
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Resumed internship after study break. Conducted thorough exploratory data analysis (EDA) on the loan dataset, examining distributions of key variables such as annual income, loan amount, debt-to-income ratio, interest rate, and repayment history. Identified class imbalance in the default label and planned mitigation strategies. Engineered features including payment compliance ratio, days since last payment, revolving utilization, and credit line metrics. Set up the data preprocessing pipeline including normalization, missing value imputation, and encoding of categorical variables.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	5th Jan 2026 – 18th Jan 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Loan Default Prediction – Model Development & Training
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Developed and trained multiple classification models for loan default prediction including Logistic Regression, Random Forest, XGBoost, and a Neural Network baseline. Implemented cross-validation and hyperparameter tuning using GridSearchCV. Addressed class imbalance using SMOTE oversampling and class-weight adjustment. Tracked evaluation metrics including accuracy, precision, recall, F1 score, and AUC-ROC. Initial results showed XGBoost achieving the highest AUC-ROC. Began building the model inference pipeline for real-time default risk scoring.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	19th Jan 2026 – 1st Feb 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Loan Default Prediction – Risk Dashboard & Project Completion
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Built the interactive Loan Default Prediction System dashboard featuring Overall Loan Default Risk Assessment. Implemented a UI with sections for Personal Information (Annual Income, Debt-to-Income Ratio), Loan Details (Loan Amount, Term, Interest Rate, Monthly Installment), Credit Information (Open/Total Credit Lines, Revolving Balance, Utilization), Repayment History, and Payment Information. Integrated the trained model to display a Default Risk Score gauge (0–100) and risk classification (Low/Medium/High Risk) with default probability percentage. Completed and documented the end-to-end project and presented findings to the mentor.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	2nd Feb 2026 – 15th Feb 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	AWS Transcribe & Sentiment Analysis – Project Initiation & Architecture Design
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Commenced the second project focused on automated call centre audio analysis using AWS cloud services. Studied the business requirement: transcribing agent-customer call recordings stored in Amazon S3 and performing NLP-based sentiment analysis. Designed the end-to-end pipeline architecture involving Amazon S3 for audio storage, AWS Transcribe for speech-to-text with speaker diarization, AWS Lambda for orchestration, and Amazon DynamoDB for storing structured output records. Set up the AWS environment, configured IAM roles and policies, created the S3 bucket (audio-analysis-input), and tested basic Transcribe API calls.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	16th Feb 2026 – 1st Mar 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	AWS Transcribe & Sentiment Analysis – Transcription Pipeline & Speaker Diarization
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Implemented the transcription pipeline with speaker diarization to separate agent (spk_0) and customer (spk_1) dialogue from audio recordings. Configured AWS Transcribe jobs with diarization enabled and parsed the diarized transcript JSON output to extract structured turn-by-turn conversation segments with speaker roles, text content, and timestamps . Stored raw transcription results in Amazon S3 and processed structured records into DynamoDB with fields including record_id, filename, s3_bucket, s3_key, language, timestamp, and diarized_transcript array. Validated output against sample call recordings.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	2nd Mar 2026 – 15th Mar 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	AWS Transcribe & Sentiment Analysis – NLP Sentiment Scoring & DynamoDB Integration
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Integrated VADER and transformer-based sentiment analysis to score both agent and customer utterances independently. Computed sentiment scores (Positive, Negative, Neutral, Compound) at the utterance level and aggregated them into overall_sentiment (POSITIVE/NEGATIVE/NEUTRAL) and overall_compound_score fields. Implemented agent_tone classification (Standard/Escalated) and customer_tone classification (Critical/Urgent/Neutral) based on compound score thresholds. Stored all results in DynamoDB including customer_scores, agent_scores, customer_satisfaction (numeric), translation_used flag, and company field. Verified end-to-end pipeline with test.mp3 sample recording showing correct sentiment extraction.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	16th Mar 2026 – 29th Mar 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	AWS Transcribe & Sentiment Analysis – Multi-language Support & Project Finalization
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Extended the pipeline to support multi-language call recordings using Amazon Translate for non-English audio. Implemented the translation_used boolean flag and populated agent_translated_text and customer_translated_text fields in DynamoDB for translated records. Conducted end-to-end testing across multiple audio samples covering Hindi and English conversations. Verified DynamoDB record structure with all required fields populated correctly. Finalized pipeline error handling, Lambda timeout configurations, and cost optimization settings. Documented the complete architecture and prepared a project summary report. Conducted demo walkthrough of the system with mentor using sample DynamoDB output.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	30th Mar 2026 – 12th Apr 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Salesforce CRM Integration – Project Initiation & Requirements Gathering
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Commenced the third project focused on Salesforce CRM integration for the Support Division. Studied the business requirements including automating customer data synchronisation, loan account management, and support case tracking within Salesforce. Set up the Salesforce Developer Org, explored the Salesforce platform architecture including Objects, Flows, Apex, and Lightning Web Components. Reviewed existing customer data schemas and mapped loan account and interaction data fields to Salesforce standard and custom objects. Configured connected app settings and OAuth 2.0 credentials for API-based integration with internal systems.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	13th Apr 2026 – 26th Apr 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Salesforce CRM Integration – Data Modelling, Flows & Apex Development
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Designed and implemented custom Salesforce objects for Loan Accounts, Customer Profiles, and Support Cases. Built Salesforce Flows for automated case creation upon customer service escalation triggers. Developed Apex classes for batch processing of loan status updates from the internal core banking system. Created Lightning Web Components for displaying customer loan summary and risk indicators on the Account page layout. Implemented validation rules and approval processes for loan modification requests. Configured Role Hierarchy and sharing settings to control data visibility across the Support Division teams. Tested all components in the sandbox environment.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

STUDENT'S DAILY DIARY/ DAILY LOG



Mukesh Patel School of Technology Management & Engineering

STUDENT'S DAILY DIARY/ DAILY LOG

Date	27th Apr 2026 –2 nd May2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Support Division, Kotak Mahindra Prime Limited (Hybrid)
Name of the finished task/product	Salesforce CRM Integration – Reporting, Dashboards & Internship Completion
Name of the External Mentor (with email id)	Nilesh Pabuwal - Nilesh.Pabuwal@kotak.com
Built Salesforce Reports and Dashboards to provide real-time visibility into loan default risk trends, open support cases by priority, and customer satisfaction scores from the sentiment analysis pipeline. Integrated the AWS sentiment analysis output with Salesforce by mapping DynamoDB records to Salesforce Case records via REST API calls in Lambda. Conducted User Acceptance Testing (UAT) with the Support Division team and resolved feedback items. Documented the full Salesforce implementation including deployment steps, configuration guide, and test coverage report. Completed final internship presentation summarising all three projects: Loan Default Prediction, AWS Transcribe & Sentiment Analysis, and Salesforce CRM Integration. Submitted all project deliverables and received mentor sign-off.	

A handwritten signature in black ink, appearing to read 'Nilesh Pabuwal'.

Signature of External Mentor

Intern Supervisor Evaluation Questionnaire

- Name of industry supervisor – NILESH PABUWAL
- Name of organization/industry and place – KOTAK MAHINDRA PRIME LIMITED

Thank you for taking the time to provide feedback on the performance of our student. This evaluation is an important part of the internship process and will be used to assess the intern's strengths, areas for improvement, and overall contribution to the team. Your insights will not only help the intern grow professionally but also assist the college in understanding the value of the internship experience. Please provide thoughtful and constructive feedback based on your observations, as this will guide the intern's development and contribute to their academic and professional progress. *(Please tick ✓ any one option)*

1. The intern demonstrates strong critical thinking skills when analysing situations and making decisions.
☐ Very poor ☐ Poor ☐ Average ☒ Good ☐ Excellent
2. The intern effectively applies management principles and concepts learned to real-world scenarios.
☐ Never applies ☐ Rarely applies ☐ Sometimes applies
☐ Usually applies ☒ Always applies
3. The intern approaches complex problems logically and finds effective solutions in a timely manner.
☐ Very Ineffective ☐ Ineffective ☐ Neutral ☐ Effective ☒ Very Effective
4. The intern communicates ideas, feedback, and information clearly and professionally in both written and verbal formats.
☐ Very Poorly ☐ Poorly ☐ Adequately ☐ Well ☒ Very Well
5. The intern demonstrates a strong desire to learn, seek feedback, and improve their skills and knowledge.
☐ Not at all interested ☐ Slightly interested ☐ Moderately interested
☒ Very interested ☐ Extremely interested
6. The intern collaborates effectively with team members, contributing positively to group dynamics and shared goals.
☐ Very Poor Collaborator ☐ Poor Collaborator ☐ Neutral
☐ Good Collaborator ☒ Excellent Collaborator

7. The intern takes proactive steps to complete tasks and find solutions without needing constant supervision.
- ☐ Never shows initiative ☐ Rarely shows initiative ☐ Sometimes shows initiative
☒ Often shows initiative ☐ Always shows initiative
8. The intern demonstrates creativity and logical thinking in designing and developing practical solutions to challenges.
- ☐ Very Poor ☐ Poor ☐ Average ☐ Good ☒ Excellent
9. The intern effectively manages their time to meet deadlines and balance competing priorities.
- ☐ Very Poorly ☐ Poorly ☐ Neutral ☐ Well ☒ Very Well
10. The intern shows the ability to adapt to new tasks, changes in the work environment, and unexpected challenges.
- ☐ Not Adaptable ☐ Slightly Adaptable ☐ Moderately Adaptable
☒ Very Adaptable ☐ Extremely Adaptable

Any other comments



Signature

Industry Project Initial Information Report (IIR)

Name of the Student	Jeet Shorey
Programme	Btech DataScience
Roll No. & SAP No.	J024-70092200086
Contact Details	
Residence	sankalp 2 a 104,poonam sagar complex ,mira road east,mumbai 401107
Mobile	9833232395
Email ID (nmims.edu.in)	jeet.shorey86@nmims.in
TIP Organization details	
Name	kotak mahindra prime ltd
Address	Kotak Infra, Building No. 21, Infocity Park, Off Western Express Highway, General Area Kumar Vastu Mang. Mahad, Mumbai 400097
City	mumbai
Pin	400097
Name of HR	nilesh pabuwai
Address of HR	Kotak Infra, Building No. 21, Infocity Park, Off Western Express Highway, General Area Kumar Vastu Mang. Mahad, Mumbai 400097
City	mumbai
Pin	400097
Contact number of HR (office)	
Contact number of HR (mobile)	9867205725
Email	nilesh.pabuwai@kotak.com
Details of the Industry Mentor	
Name	harishankar vashishtha
Designation	senior full stack developer
Contact number (office)	
Contact number (mobile)	9058583987
Email	harishankar.vashishtha@kotak.com
TIP reporting date	24-nov-2025
TIP Period	6 months
Office timings at the Organization	hybrid
Stipend expected (if any)	none
Project Title	call transcription and sentimental analysis
Project Description (in 20 words)	Kotak Bank converts customer calls into text to analyze emotions, ensuring service quality, regulatory compliance, and improved customer experience through AI.

 **jeet shorey**

Name & Signature of Student

harishankar vashishtha
Name & Signature of the Industry Mentor

Date: 17 jan 2026

